

Abdullah Mahmoud Abdullah

Location: AL-Wasta, Bani Suef | **Phone:** +201155765080 | **Email:** abduallah.mahmoud.abduallah1344@gmail.com
LinkedIn: abduallah-mahmoud-de1344 | **GitHub:** abduallah-mahmoud-datastack | **Portfolio:** abduallah-mahmoud-de.netlify.app

PROFESSIONAL SUMMARY

Data Engineer specializing in cloud-native data platforms, ETL/ELT pipeline development, and data warehousing. Experienced in building scalable solutions using AWS, Snowflake, Apache Airflow, Spark, and Python. Delivered real-world data engineering and AI-powered automation projects, including serverless pipelines, real-time streaming systems, and analytics platforms.

SKILLS

- Programming & Scripting:** Python, SQL, Bash/Shell Scripting
- Cloud Platforms:** AWS Solutions Architecture, GCP (BigQuery), Snowflake
- Data Engineering & Big Data:** ETL/ELT Pipelines, Apache Spark, Hadoop, Apache Airflow, Apache Kafka
- Databases:** OLTP (PostgreSQL, RDBMS), OLAP & Data Warehousing, NoSQL
- Business Intelligence & Visualization:** Looker Studio, Streamlit, IBM Cognos Analytics
- Machine Learning & AI:** Generative AI (Google Gemini, Groq API), Predictive Modelling, NLP (Whisper V3)
- Soft Skills:** Structured Problem Solving

EXPERIENCE

Roto House | AI Engineer | Remote Oct 2025 – Present

- Developed a factory management web app with an AI-driven backend (Node.js, PostgreSQL for OLTP) to automate reporting and digitize industrial workflows, streamlining operations for 70+ active technicians.
- Processed 20+ daily audio using FFmpeg and Whisper V3 (via Groq API) for accurate speech-to-text transcription.
- Reduced manual data entry time by 60% by leveraging Google Gemini to analyze visual workflows and extract key operational data, successfully auto-populating 12+ target forms daily.

PROJECTS

IMDb Data Pipeline (AWS → Snowflake)

- Built an automated AWS ELT pipeline using EventBridge, Lambda, and SNS to extract over 670 MB IMDb datasets.
- Optimized Snowflake OLAP ingestion using CDC and SQL MERGE, eliminating full-loads to achieve high data availability.
- Developed a Streamlit dashboard powered by Snowflake, enabling live data visualization via SQL pushdown.

Real-Time Crypto Streaming Pipeline

- Engineered an event-driven AWS pipeline (SQS, Lambda, DynamoDB) to process live Binance WebSocket data.
- Provisioned infrastructure and implemented real-time automated SNS alerts for micro-batch market volatility.
- Built a Streamlit dashboard integrating Gemini 3-flash-preview for live quantitative market analysis.

Enterprise Data Engineering Platform

- Architected OLTP (relational) and NoSQL databases to handle diverse data types securely and efficiently.
- Built and orchestrated automated ETL pipelines using Python, Bash, and Apache Airflow, reducing manual processing.
- Processed large-scale datasets using Apache Spark, integrating the data warehouse with Looker Studio for interactive dashboards.

YouTube Analytics Pipeline

- Engineered a data pipeline, utilizing AWS Lambda for the extraction of raw JSON and CSV files from Amazon S3.
- Processed and optimized datasets using AWS Glue by partitioning data and converting files to Parquet format.
- Centralized the transformed data within Snowflake and developed dashboards using Looker Studio and Streamlit.

Weather Analytics ETL Pipeline

- Engineered an automated Python and Airflow ETL pipeline to extract daily weather data from the Open-Meteo API.
- Transformed complex, nested JSON payloads into structured, analysis-ready datasets utilizing Pandas.
- Designed an interactive Streamlit dashboard for dynamic visualization and analysis of global weather trends.

CERTIFICATIONS

- IBM Data Engineering Professional Certificate** | Coursera | Mar 2026
- Snowflake Data Engineering Professional Certificate** | Coursera | Apr 2026
- AWS Cloud Solutions Architect** | Coursera | May 2026

EDUCATION

Bachelor of Computer Science | Faculty of Computers and Artificial Intelligence | Sep 2022 - Jul 2026

- Relevant Coursework:** Data Modeling, Data Architecture, Database Management, Artificial Intelligence.